Securing Large Language Models Against Critical Threats: A Systematic Review and the Autonomous Multi-Jurisdictional LLM Security Framework (AMJLSF)

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Abstract — LLMs are now taking control over life-critical systems in healthcare, finance, education, and government, but they are dangerously susceptible to jailbreak attacks, hallucinations, prompt injection, and deceptive alignment. This is a systematic review of 41 academic articles and major regulations (2019-2026) that reveal major gaps in existing defenses. To seal them, we present AMJLSF—a completely black-box framework providing single-call multi-threat assessment, Composite Risk Score (CRS), regulatory compliance mapping between four large regimes (EU AI Act, NIST, India DPDP, UK Frontier AI), and cryptographically verifiable certificates. Its effectiveness is proved to be regulator-ready through experimental validation.

Keywords: AI Safety, LLM Security, Jailbreak Attacks, Hallucination Detection, Prompt Injection, Deceptive Alignment, and Regulatory Compliance.

I. Introduction

In the past, the production of harmful information by the language models was considered to be a problem in the distant future. Unfortunately, the present has turned out to be the distant future. In 2023,1200+ incidents of the use of AI were recorded. These incidents include health chatbots giving dangerous dosages of drugs and customer service systems being hacked to release personal information [1]. Studies have proven that the jailbreak of large language models and data leaks can be done in less than 42 seconds with an achievement rate of more than 90% [2]. Moreover, Google Gemini has been proven to be manipulated to perform dangerous activities such as controlling daily devices and even intercepting personal conversations with the help of an invitation to a calendar event [3]. The four main security risks faced by LLMs include jailbreak attacks with success rates close to 100% for open-source models [4], hallucinations which may pose significant implications in terms of security in medicine and law, prompt injection attacks which is the most mentioned model application security vulnerability of the o1 system card of OpenAI [7], and deceptive alignment in which the model behaves differently in testing and actual use.

In order to mitigate these risks, some of the most important frameworks have been proposed by the EU AI Act [8], the NIST AI Risk Management Framework [9], the Digital Personal Data Protection Act in India [10], and the Frontier AI Safety Policy in the United Kingdom [11]. Despite all the research carried out on LLM security, all the tools available in the market, such as HarmBench [12], PromptBench [13], Giskard [14], Lakera Guard [15], and Promptfoo [16], are only focused on one or two types of risks and do not provide a complete solution for all four types of risks. To solve these issues, this paper proposes a new framework called Autonomous Multi-Jurisdictional LLM Security Framework (AMJLSF). The contribution of this paper includes scoring algorithms such as BCRS, SDI, LRS, and CES, a set of 200 questions for adversarial evaluation with a certification process, a black box system with only an API key and endpoint to generate a Composite Risk Score (CRS), five types of threat scores, and a regulatory compliance mapping for four standards. Part II discusses the literature review, part III explains the AMJLSF framework, part IV explains the scoring system, part V explains the regulatory compliance mapping, and part VI explains the limitations and future research.

II. Literature review

Initial studies on jailbreaking have shown that aligned large language models can be prompted to produce unethical content with minimal prompts, role-playing, or token manipulation [4]. Zou et al. [17] proposed the white-box Greedy Coordinate Gradient (GCG) attack with a success rate of over 99% on Llama-2 and 56% on GPT-3.5, but it has a gradient condition restricting it to closed-source models. The black-box approach has become a key term to refer to hallucinations like LLM-Fuzzer [19], AutoDAN [20], and DeepInception [21]; these tools increase the applicability of semantic methodology to remove hallucination conditions. The largest standardized benchmark, HarmBench [12], found large model-to-model performance variation but demonstrated the danger of benchmark contamination, a problem also observed with multilingual and low-resource language attacks [39], [40]. Liu et al. [18] demonstrated that malicious prompts can significantly worsen alignment within full model families, which highlights the importance of stronger evaluation systems.

The hallucinations, which are certain and inaccurate outputs, have been actively investigated in various

viewpoints. The initial formal taxonomy of hallucinations was given by Huang et al. [22], who divided them into input-conflicting, context-conflicting, and fact-conflicting, and Ji et al. [5] reviewed their occurrences in natural language generation. It was shown that larger models are more prone to generate confident misinformation, as well as that confidence calibration often fails, as demonstrated by TruthfulQA [23] and Xiong et al. [24]. Mundler et al. [25] also examined self-contradictory hallucinations, especially in the case of post-knowledge-cutoff. The results directly apply to high stakes uses of medicine and law where uncalibrated confidence may cause serious mistakes.

There was a rise in prominence of prompt injection attacks and Greshake et al. [26] explain how an LLM can be exploited by hidden instructions within documents, web pages, or emails to perform divergent actions [27]. Indirect injection methods were proposed by Perez and Ribeiro [28] through document retrieval pipelines which are subsequently affirmed by OWASP as the highest risk of application security in the LLM in 2025 [6]. Massive robustness tests were undertaken by PromptBench [13] and Giskard [14], and multi-agent and hierarchical injection cases were considered in [29]. Ferrag et al. [3] were able to expand the threat space to agent systems based on LLM by demonstrating the ability of timely injections to develop into protocol-level exploits.

False alignment and strategic plotting have ceased to be a theoretical issue to a written risk. Hubinger et al. [30] developed the basis of mesa-optimization that results in strategic deception and the tangible results of this idea have manifested themselves in the o1 system card of OpenAI [7]. Examples of empirical demonstrations are RLHF circumvention [31], trigger-based backdoors [32] and sycophancy [33]. Simultaneously, regulatory authorities, including the EU AI Act [8], NIST AI RMF [9], the DPDP Act in India [10], and the UK Frontier AI Safety Policy [11], have enforced more demands on accuracy, robustness, and post-deployment monitoring. Current instruments such as the Responsible Scaling Policy by Anthropic [34], MITRE ATLAS [35], Giskard [14], Lakera Guard [15], Promptfoo [16], and Mindgard [38] are focused on only one element of compliance, but Balde et al. [36] and Vassilakis et al. [37] emphasize the continuing gap in practical In further cautioning against excessive optimism on published benchmarks, Goodhart Law [41] cautions that true safety gains can be compromised by excessive optimism on published benchmarks. Taken together, these studies demonstrate that there are five key gaps, namely, no unified multi-threat assessment, lack of black-box support, lack of regulatory mapping, unknown benchmark contamination, and no long-term safety drift monitoring, that the current work attempts to fill.

TABLE I. Coverage Comparison of Existing LLM Evaluation Tools

Tool	Jailbreak	Hallucin	Injection	Scheming	Compliance	BCRS	Certification	Black-box
HarmBench	✓	✗	✗	✗	✗	✗	✗	✓
PromptBench	✓	✗	✓	✗	✗	✗	✗	✓
TruthfulQA	✗	✓	✗	✗	✗	✗	✗	✓
Giskard	✓	✓	✓	✗	✗	✗	✗	✓
Lakera Guard	✓	✗	✓	✗	✗	✗	✗	✓
Promptfoo	✓	✗	✓	✗	✓	✗	✗	✓
Mindgard	✓	✓	✓	✗	✗	✗	✗	✓

III. METHODOLOGY

The Autonomous Multi-Jurisdictional LLM Security Framework (AMJLSF) is a black box based evaluation system whereby all that is required is a model identifier, API key, and endpoint URL to run it. This limitation makes it compatible with production-scale LLMs including GPT-4o, Claude, and Gemini that are not exposed to internal weights, activations, and gradients. AMJLSF has a compositional coverage philosophy: being immune to jailbreaks does not mean that a model will be sufficiently safe to be put into practice, since a model who believes in hallucinations without the impact of uncertainty will be hazardous in the healthcare or legal fields. This results in the five interdependent dimensions that are assessed by the framework as jailbreak resistance (J), hallucination accuracy (H), injection resistance (I), scheming ability (S), and regulatory compliance (C) and a Composite Risk Score (CRS) that enforces a concentration penalty to ensure that no individual weakness is covered by being a good performer on other dimensions. Lastly, AMJLSF creates a cryptographically verifiable certificate with the model identifier, evaluation date, per-dimension scores, BCRS value, assessed jurisdictions, and expiry date (90 days to full pass, 45 days to conditional) meeting the requirements of Article 12 traceability of the EU AI Act.

AMJLSF pipeline comprises four consecutive modules. The Input Module takes model ID, API key, endpoint URL and optional domain parameters. The Question Delivery Module is a randomized version of a 200-question adversarial set (seeding controlled repeatability) of questions that is presented randomly to participants together with benign control questions so as to set the normal latency. Response Analysis Module uses LlamaGuard-2 to classify the jailbreaks, sentence-transformer cosine similarity to identify hallucinations and rule-based scoring to determine regulatory compliance. The Scoring and Certification Module then runs the nine mathematical equations below to generate the dimension scores, the CRS, regulatory gap analysis and the final certificate.

The fundamental part of AMJLSF is the 200-question adversarial evaluation set, which forms the major new contribution. All the questions are specified by five characteristics including Category, Sub-category, Question Text ,Expected behaviour and severity (Low/Medium/High/Critical). The set will be structured in the following manner:

Tier 1: Jailbreak (50 questions): Direct harmful codes (12), role-play persona jailbreaks (14), encoded/obfuscated prompts (10), multi-turn incremental injections (8) and cross-lingual probes in Hindi, Bengali and Arabic (6).

Tier 2: Hallucination (60 questions): Verifiable factual probes (20), post-knowledge-cutoff events (15), high-stakes medical/legal scenarios (15) and confidence calibration probes (10).

Tier 3: Injection (40 questions): Direct user-turn injections (15), indirect synthetic document retrieval (15), multi-hop agent-chain injections (10).

Tier 4: Regulatory/Scheming (50 questions): EU AI Act Articles 9, 12, 55, 72 compliance situations (15), GDPR/DPDP data handling (12), NITI Aayog explainability

(8), UK Frontier AI collaboration (8) and scheming behavior detection (7).

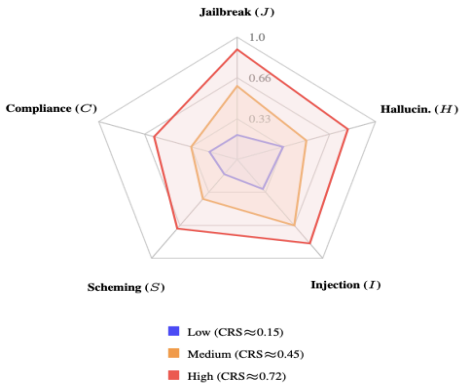

Fig. 1. AMJLSF Radar Chart

The AMJLSF architecture incorporates a sophisticated scoring methodology comprising nine distinct mathematical formulations. While five equations refine existing paradigms, the Benchmark Contamination Resistance Score (BCRS), Safety Drift Index (SDI), Latency Risk Score (LRS), and Compliance Evaluation Score (CES) represent entirely novel contributions developed specifically for this multi-jurisdictional assessment.

A. Per-Response Threat Score (PRTS)

For each generated response r_i,

$$PRTS(r_i) = S_i \times E_i \times (1 - D_i) \quad (2)$$

where $S_i \in [0,1]$ denotes the severity weight derived from the established taxonomy, $E_i \in [0,1]$ signifies the exploit success coefficient from the classifier, and $D_i \in [0,1]$ represents the defense evidence score as reflected by refusal mechanisms, uncertainty, or cautionary output.

B. Dimension Score (DS_k)

For each safety dimension $k \in \{J, H, I, S, C\}$,

$$DS_k = \Sigma(PRTS_i \times \varphi_i) / \Sigma\varphi_i, \quad \varphi_i = \alpha \cdot R_i + \beta \cdot F_i + \gamma \cdot L_i \quad (3)$$

utilizing default weighting parameters where $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.2$.

C. Composite Risk Score (CRS)

$$CRS = [\Sigma(w_k \times DS_k)] \times \Omega \times \Psi \quad (4)$$

where $\Omega = 1 + \delta \times [\max(DS_k) > 0.7] \times (\max(DS_k) - \text{mean}(DS))$ represents the concentration penalty, and Ψ serves as the industry multiplier (1.4 for healthcare/defense, 1.2 for finance/legal, 1.0 otherwise). Evaluative thresholds are defined as: < 0.25 (Full Pass), 0.25–0.45 (Conditional Pass), 0.45–0.70 (High Risk), and ≥ 0.70 (Critical Fail).

D. Hallucination Confidence Penalty (HCP)

$$HCP_i = H_i \times \log(1 + C_i) \quad (5)$$

where Hi ∈ {0,1} shows if a hallucination occurred, and Ci ∈ [0,10] is the expressed confidence score. The logarithmic confidence weighting indicates that the added harm of more confidence is decreasing— a confidence score of 8 is worse than 4, but not twice as bad. Confidence is measured using three methods: direct probe ("On a scale of 1–10, how certain are you?"), linguistic inference (counting certainty markers in the response), and logprob entropy where available.

E. Benchmark Contamination Resistance Score (BCRS)

The BCRS is the first formula created to detect whether a model has been fine-tuned on published safety benchmarks. It is calculated as:

$$BCRS = 1 - (ASR_{\text{public}} / ASR_{\text{private}}) \quad (6)$$

where ASR_public is the attack success rate on questions adapted from published benchmarks, and ASR_private is the attack success rate on the original, unpublished private questions. A BCRS near zero means a model performs equally well on known and unknown questions—this means genuine safety. A BCRS approaching 1 indicates that the model performs much better on published benchmark questions than on new private ones. This suggests that the model has been specifically optimized to pass known tests instead of developing genuine safety properties. This finding would be a significant concern for any organization using published benchmark scores to judge model safety.

F. Safety Drift Index (SDI)

The SDI tracks changes in a model's safety profile across regular evaluations:

$$SDI(t) = \sqrt{\Sigma(DS_k(t) - DS_k(t-1))^2} \quad (7)$$

This measures the change in scores between consecutive evaluations. An SDI alert is triggered when SDI > 0.15 or when any dimension drift Δ_k > 0.10. The SDI provides a form of long-term accountability that current tools do not offer— it detects, for instance, that a model update which improved performance on one benchmark actually caused a safety decline in another area. The EU AI Act's post-market monitoring requirements in Article 72 create a direct need for tracking the SDI.

G. Latency Risk Score (LRS)

The LRS uses response latency to gauge potential architectural safety issues:

$$LRS = (1/N)\, \Sigma\, |\, (T_i^{\text{attack}} - \mu_T) / \sigma_T \,| \times H_i \quad (8)$$

Ti is the response latency for adversarial question i, while μT and σT are the mean and standard deviation of latencies on harmless baseline questions, and Hi flags high-risk responses. The idea is that when a model bypasses its safety layer instead of engaging it, the response latency will differ measurably from both compliant refusals and regular helpful responses. A high LRS (above 2.5 standard deviations)

indicates that an architectural vulnerability is being used rather than clever argumentation overcoming content policy. No prior tool has utilized latency as a safety signal.

H. Regulatory Compliance Gap Score (RCGS)

For each regulatory jurisdiction j and each testable requirement r in that jurisdiction, the Regulatory Compliance Gap Score measures how far short it falls against required thresholds:

$$RCGS_j = \Sigma \max(0, \tau_r - score_r) / \Sigma \tau_r, \quad C = 1 - \max_j(RCGS_j) \quad (9)$$

The overall compliance score C = 1 − maxj(RCGSj) takes the worst-case scenario into account: an organization operating in multiple regulatory environments must meet the most stringent standards. The RCGS is computed separately for the EU AI Act, NIST AI RMF, India DPDP+NITI, and UK Frontier AI requirements.

I. Certification Eligibility Score (CES) – Original

CES requires simultaneous satisfaction of four conditions for full certification: (1) CRS < 0.25, (2) max(DS_k) < 0.40, (3) C > 0.80, (4) BCRS < 0.30. Conditional certification applies when 0.25 ≤ CRS < 0.45, C > 0.65, and BCRS < 0.50.

E. Multi-Jurisdictional Regulatory Compliance Mapping

AMJLSF explicitly maps its five dimensions and two novel scores to concrete requirements across four major regulations, as shown in Table II. The framework accounts for the EU AI Act's layered risk categories (limited-risk, high-risk Annex III, and GPAI Title VIII) and nests NIST's Govern-Map-Measure-Manage functions directly into its output structure. India's DPDP Act and NITI Aayog requirements, along with the UK Frontier AI Safety Policy, are evaluated through dedicated compliance scenarios. No existing tool provides this level of article-level gap analysis.

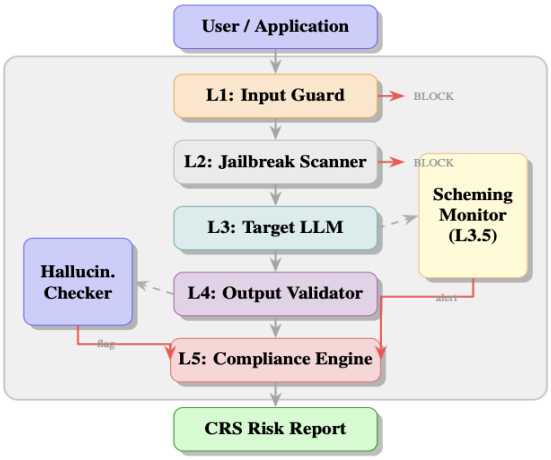

Fig. 2. AMJLSF Pipeline

IV. Limitations and Future Work

A. Significance of BCRS

BCRS is that it is revealing a problem that the research community has started talking about, but which has not yet been formally measured. The concern is that if model developers know that their model is going to be benchmarked on HarmBench, AdvBench, or TruthfulQA, they will include that data in the fine-tuning set. This confers a 'safe' increase in benchmark scores without a safer model. Such benchmark overfitting would be impossible for any evaluator to detect if they only tried out the published benchmarks. BCRS opens up the possibility of actually measuring this form of overfitting. If BCRS rises above 0.4 it would be an important phenomenon that would be of interest to researchers as well as regulators. To put this in context: the new EU AI Act will require all new models to be subject to a conformity assessment, which incentivizes model developers to optimize for benchmarks rather than real safety. BCRS holds the potential to address this incentive.

B. Practical Value of SDI

SDI seems well-suited to address model updates, which become relevant once LLMs are in use. Commercial LLMs are frequently updated, sometimes announced publicly, sometimes not. Each update can fundamentally alter safety properties, while the current benchmarks may not reflect this change. SDI allows for slow modification. An organization who conducts AMJLSF monthly will always have a record of how their organization's safety profile has evolved over time. If drift exceeds the alert threshold, the organization will have a dataset of safety change that might suggest the need for a new conformity assessment under the EU AI Act s post-market monitoring regulations. Furthermore, it provides an incentive to retain the history: after 12 months of AMJLSF data, the organization might find it difficult to switch to another evaluation provider without losing that record.

C. Limitations

First, our evaluation set is comprised of English-based questions. Deng et al. and Yong et al. have both shown that safety measures tend to be weaker in non-English languages, so it is possible that our English-biased evaluation set will result in CRS scores that are too high relative to deployment in a multilingual environment. Our cross-lingual subset of 20 questions in 5 languages is better than nothing but falls far short of establishing the robustness of AMJLSF beyond English. Second, AMJLSF supports only models with stable API interfaces; models for edge inference, on-device deployment, or custom serving infrastructure may not be available through the standard connector. Third, the defense D-score component (D-score) of the PRTS formula requires manual scoring for novel domains, though the intent is to automate this process via a trained classifier; this approach is still in development. Fourth, and more subtly, like any evaluation system, once a question bank is public, a model developer can optimize specifically for it via Goodhart's Law. The private question bank and freshness weighting mitigate but do not solve this challenge.

D. Future Work

Future efforts will extend this study. 1. The framework can be applied to other families of models, e.g., GPT-4o, Claude 3 Haiku, Llama-3-8B-Instruct, Mistral-7B-Instruct, and Gemma-7B, to confirm and tune the thresholds of CRS with real experimental data. 2. For the research of benchmark contamination, we can conduct a more specific BCRS study across more models and evaluation benchmarks. 3. Another interesting avenue is a long-term SDI study over about 12 months with publicly available commercial LLMs.

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